IT 314 Assignment 2: Multi-client Chat Server

This assignment asks you to build a small chat system that many clients can use at the same time. Read the whole brief before you start.

1. Functional requirements

1. The server must accept at least 10 clients connected at the same time.
2. Each client must choose a unique nickname when it connects.
3. A message sent by one client must be delivered to every other connected client.
4. The server shall announce when a client joins or leaves the chat.
5. Clients should be able to send a private message with the command /msg .
6. The server may keep a history of the last 50 messages for new clients.

2. Quality requirements

- The server must not crash when a client disconnects without warning.
- Messages should reach other clients within one second on a local network.

3. Constraints

- Use Python 3.11 and the standard library only.
- Use TCP sockets; do not use a web framework.
- Work in groups of at most two students.

4. What to submit

- A ZIP file with the source code of the server and the client.
- A README that explains how to run both programs.
- A two-page report describing your design and how you tested it.

Submit everything on the course portal before 23:59 on Friday of week 6.

5. Marking

Criterion	Marks
Correct message broadcasting	30
Nicknames and join/leave notices	15
Private messages	15
Robustness to disconnects	20
Code quality and README	10
Design report	10

Total: 100 marks

Late submissions lose 10 % per day. This note is information, not a requirement for your program.